

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
# Loading packages
library(tidyverse)
library(lubridate)
library(here)

# Get file path
getwd()
```

```
## [1] "/home/guest/EDE_FALL 2025"
```

```
# Read or get file
Neonics <- read.csv(
  file = here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),
  stringsAsFactors = TRUE)

# Read or get second file: Litter data
Litter <- read.csv(
  file = here("Data/Raw/NEON_NIWOLitter_massdata_2018-08_raw.csv"),
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Based on the dataset and information online, we are interested in the ecotoxicology of neonicotinoids because they are a insecticide designed to kill pest insects in agriculture and gardens. However, their use can harmful consequences for other insects and bugs.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Studying forest litter is crucial because it's a key part of nutrient cycling. Which the litter would turn into organic matter that can help support soil health and other small organisms for growth.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Different trap and in sizes: litter is using elevated 0.5m² PVC traps while fine woody debris is collected using ground traps. 2. Sample frequency: elevated traps are sampled every two weeks and ground traps are sampled per year. 3. Collected sample is sorted into functional groups and weight, masses are then reported per trap per collection event.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

```
dim(Litter)
```

```
## [1] 188 19
```

```
# The dimensions of Neonics data is 4623 observations and 30 variables  
# The dimensions of Litter data is 188 observations and 19 variables
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
effect_summary <- summary(Neonics)  
sort(effect_summary, decreasing = TRUE)
```

```
## [1] "Year(s) : 64 "  
## [2] "Williamson,S.M., S.J. Willis, and G.A. Wright : 93 "  
## [3] "Whole organism : 41 "  
## [4] "Unmeasured values (some measured values reported in article): 16 "  
## [5] "Unmeasured :4321 "  
## [6] "Toxicity of Neonicotinoid Insecticides on Different Honey Bee Genotypes  
## [7] "Topical, general : 254 "  
## [8] "Tiller stage : 15 "  
## [9] "Technical grade, technical product, technical formulation: 422 "  
## [10] "Survival : 133 "  
## [11] "Spray : 393 "  
## [12] "Scholer,J., and V. Krischik : 82 "  
## [13] "Reproduction : 197 "  
## [14] "Reduced Risk Insecticides to Control Scale Insects and Protect Natural Enemies in the Product.  
## [15] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Formul  
## [16] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active  
## [17] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active  
## [18] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active  
## [19] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active  
## [20] "Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active  
## [21] "Pupa : 69 "  
## [22] "Progeny counts/numbers: 120 "  
## [23] "ppm : 180 "  
## [24] "Population :1803 "  
## [25] "Popillia japonica : 94 "  
## [26] "PLoS One9(3): 14 p. : 82 "  
## [27] "Pestanal grade : 93 "  
## [28] "Peck,D.C. : 208 "  
## [29] "Parasitic Wasp : 285 "  
## [30] "NR/ : 108 "  
## [31] "NR-LETH: 86 "  
## [32] "NR :3851 "  
## [33] "NR :2502 "  
## [34] "NR : 355 "  
## [35] "NR : 217 "  
## [36] "NR : 167 "  
## [37] "NR : 94 "  
## [38] "Not reported:2271 "
```

```

## [39] "Not reported: 663 "
## [40] "Not reported      :3515 "
## [41] "Not reported      :4349 "
## [42] "Not reported      :3989 "
## [43] "Not reported      : 5 "
## [44] "Not coded : 51 "
## [45] "Not coded : 42 "
## [46] "Not coded : 51 "
## [47] "Not coded : 53 "
## [48] "Not coded : 51 "
## [49] "NOEL :1816 "
## [50] "No substrate:2934 "
## [51] "ng/org : 231 "
## [52] "Natural soil: 393 "
## [53] "N''-Methyl-N-nitro-N'-[(tetrahydro-3-furanyl)methyl]guanidine : 218
## [54] "Multiple : 285 "
## [55] "Mortality :1493 "
## [56] "Mortality :1294 "
## [57] "Min. :1982 "
## [58] "Min. : 58842209 "
## [59] "Min. : 344 "
## [60] "Midgut or midgut gland: 63 "
## [61] "Median :2010 "
## [62] "Median :165559 "
## [63] "Median :138261413 "
## [64] "Measured : 230 "
## [65] "Mean :2008 "
## [66] "Mean :147651982 "
## [67] "Mean :142189 "
## [68] "Max. :210880925 "
## [69] "Max. :2019 "
## [70] "Max. :180410 "
## [71] "Long-Term Effects of Imidacloprid on the Abundance of Surface- and Soil-Active Nontarget Fauna
## [72] "LOEL :1664 "
## [73] "Litter : 264 "
## [74] "LD50 : 274 "
## [75] "LC50 : 327 "
## [76] "Laurino,D., A. Manino, A. Patetta, and M. Porporato : 88 "
## [77] "Larva : 437 "
## [78] "Lab :2860 "
## [79] "Italian Honeybee : 113 "
## [80] "Instar : 255 "
## [81] "Insects/Spiders; U.S. Invasive Species : 360 "
## [82] "Insects/Spiders; Standard Test Species; U.S. Invasive Species: 667 "
## [83] "Insects/Spiders; Standard Test Species : 27 "
## [84] "Insects/Spiders :3569 "
## [85] "Hypopharyngeal gland : 27 "
## [86] "Hours post-emergence: 99 "
## [87] "Hour(s) : 241 "
## [88] "Honey Bee : 667 "
## [89] "Head : 23 "
## [90] "Hand spray : 210 "
## [91] "Growing season : 48 "
## [92] "Ground granular : 249 "

```

```

## [93] "Frank,S.D.                                     : 100  "
## [94] "Formulation           :1420  "
## [95] "Food consumption      : 103  "
## [96] "Food                  :1124  "
## [97] "Filter paper: 230  "
## [98] "Field undeterminable:  4  "
## [99] "Field natural         :1663  "
## [100] "Field artificial     :  96  "
## [101] "Feeding behavior: 255  "
## [102] "Exposure to Neonicotinoids Influences the Motor Function of Adult Worker Honeybees
## [103] "Environmental, unspecified:1599  "
## [104] "Environ. Entomol.41(2): 377-386                      : 100  "
## [105] "Emergence           :  70  "
## [106] "Emergence           :  98  "
## [107] "El Hassani,A.K., M. Dacher, V. Gary, M. Lambin, M. Gauthier, and C. Armengaud:  96  "
## [108] "Egg                 : 128  "
## [109] "Effects of Sublethal Doses of Acetamiprid and Thiamethoxam on the Behavior of the Honeybee (A
## [110] "Ecotoxicology23:1409-1418                          :  93  "
## [111] "Development         : 136  "
## [112] "Day(s) post-hatch   :  20  "
## [113] "Day(s) post-emergence:  17  "
## [114] "Day(s)              : 327  "
## [115] "Day(s)              :4394  "
## [116] "Commercial grade                                :  27  "
## [117] "Chronic Exposure of Imidacloprid and Clothianidin Reduce Queen Survival, Foraging, and Nectar
## [118] "Carniolan Honey Bee : 152  "
## [119] "Bumble Bee         : 140  "
## [120] "Bull. Insectol.66(1): 119-126                      :  88  "
## [121] "Buff Tailed Bumblebee: 183  "
## [122] "Bombus terrestris   : 183  "
## [123] "Bombus impatiens    : 140  "
## [124] "Behavior            : 360  "
## [125] "Arch. Environ. Contam. Toxicol.54(4): 653-661:  96  "
## [126] "Apis mellifera ssp. ligustica: 113  "
## [127] "Apis mellifera ssp. carnica : 152  "
## [128] "Apis mellifera      : 667  "
## [129] "Analytical grade                                :  15  "
## [130] "AI mg/L           : 298  "
## [131] "AI lb/acre: 277  "
## [132] "AI kg/ha : 575  "
## [133] "AI g/ha : 241  "
## [134] "Agric. For. Entomol.11(4): 405-419                  : 200  "
## [135] "Adult             :1222  "
## [136] "Active ingredient:3161  "
## [137] "Abundance         :1699  "
## [138] "75                :  89  "
## [139] "70                : 112  "
## [140] "7                 : 207  "
## [141] "6                 : 230  "
## [142] "50                : 200  "
## [143] "5                 : 314  "
## [144] "4                 : 221  "
## [145] "4                 :  81  "
## [146] "3rd Qu.:2013  "

```

```

## [147] "3rd Qu.:168998  "
## [148] "3rd Qu.:153719234  "
## [149] "3-[(2-Chloro-5-thiazolyl)methyl]tetrahydro-5-methyl-N-nitro-4H-1,3,5-oxadiazin-4-imine: 686
## [150] "3      : 499  "
## [151] "3      : 183  "
## [152] "3      : 105  "
## [153] "25     : 244  "
## [154] "20     : 189  "
## [155] "2      :2441  "
## [156] "2      : 383  "
## [157] "2      : 111  "
## [158] "1st Qu.:2005  "
## [159] "1st Qu.:138261413  "
## [160] "1st Qu.:108459  "
## [161] "1023   : 80  "
## [162] "10/    : 127  "
## [163] "1      : 713  "
## [164] "1      : 82  "
## [165] "1      : 59  "
## [166] "0.37/  : 208  "
## [167] "0.0417 : 133  "
## [168] "<24    : 81  "
## [169] "[N(Z)]-N-[3-[(6-Chloro-3-pyridinyl)methyl]-2-thiazolidinylidene]cyanamide : 128
## [170] "[C(E)]-N-[(2-Chloro-5-thiazolyl)methyl]-N'-methyl-N''-nitroguanidine : 452
## [171] "(Other):3924  "
## [172] "(Other):2649  "
## [173] "(Other):1287  "
## [174] "(Other): 701  "
## [175] "(Other): 335  "
## [176] "(Other): 289  "
## [177] "(Other) :2821  "
## [178] "(Other) : 211  "
## [179] "(Other) : 88  "
## [180] "(Other) : 379  "
## [181] "(Other) : 122  "
## [182] "(Other) :3083  "
## [183] "(Other) : 59  "
## [184] "(Other) :1176  "
## [185] "(Other) : 69  "
## [186] "(Other) : 794  "
## [187] "(Other) :3274  "
## [188] "(Other) :3964  "
## [189] "(Other) : 24  "
## [190] "(Other) :3956  "
## [191] "(Other) : 61  "
## [192] "(Other)
## [193] "(Other)
## [194] "(2E)-1-[(6-Chloro-3-pyridinyl)methyl]-N-nitro-2-imidazolidinimine :2658
## [195] "(1E)-N-[(6-Chloro-3-pyridinyl)methyl]-N'-cyano-N-methylethanimidamide : 420

```

```

effect_summary2 <- summary(Litter)
sort(effect_summary2, decreasing = TRUE)

```

```
## [1] "Y: 20  "
```

```

## [2] "Woody material:26 "
## [3] "Twigs/branches:28 "
## [4] "szrillo@battelleecology.org:97 "
## [5] "Other          :24 "
## [6] "NIW0:188 "
## [7] "NIW0_067.basePlot.ltr:17 "
## [8] "NIW0_067:17 "
## [9] "NIW0_067_017:17 "
## [10] "NIW0_061.basePlot.ltr:17 "
## [11] "NIW0_061:17 "
## [12] "NIW0_061_169:17 "
## [13] "NIW0_058.basePlot.ltr:16 "
## [14] "NIW0_058:16 "
## [15] "NIW0_058_101:16 "
## [16] "NIW0_046.basePlot.ltr:18 "
## [17] "NIW0_046:18 "
## [18] "NIW0_046_155:18 "
## [19] "NIW0_041.basePlot.ltr:19 "
## [20] "NIW0_041:19 "
## [21] "NIW0_041_059:19 "
## [22] "NIW0_040.basePlot.ltr:20 "
## [23] "NIW0_040:20 "
## [24] "NIW0_040_205:20 "
## [25] "NEON.LTR.NIW0061169.20180802: 9 "
## [26] "NEON.LTR.NIW0058101.20180802: 9 "
## [27] "NEON.LTR.NIW0046155.20180802: 10 "
## [28] "NEON.LTR.NIW0041059.20180830.NDL: 2 "
## [29] "NEON.LTR.NIW0041059.20180830.MXT: 2 "
## [30] "NEON.LTR.NIW0041059.20180830: 11 "
## [31] "NEON.LTR.NIW0040205.20180830.NDL: 2 "
## [32] "NEON.LTR.NIW0040205.20180830.MXT: 2 "
## [33] "NEON.LTR.NIW0040205.20180830: 10 "
## [34] "NEON.LTR.NIW0040205.20180802.NDL: 2 "
## [35] "NEON.LTR.NIW0040205.20180802.MXT: 2 "
## [36] "NEON.LTR.NIW0040205.20180802: 10 "
## [37] "NEON.DOC.001710vE:188 "
## [38] "Needles          :30 "
## [39] "NA's:188 "
## [40] "N:168 "
## [41] "Mode:logical "
## [42] "Min.      :0.0000 "
## [43] "Median    :0.0050 "
## [44] "Mean      :0.6115 "
## [45] "Max.      :8.6300 "
## [46] "Leaves          :24 "
## [47] "kstyers@battelleecology.org:91 "
## [48] "Flowers          :23 "
## [49] "D13:188 "
## [50] "3rd Qu.:0.3200 "
## [51] "2018-09-05T19:30Z:97 "
## [52] "2018-09-05:97 "
## [53] "2018-08-30T22:30Z:97 "
## [54] "2018-08-30:97 "
## [55] "2018-08-06T18:02Z:91 "

```

```
## [56] "2018-08-06:91  "
## [57] "2018-08-02T21:00Z:91  "
## [58] "2018-08-02:97  "
## [59] "2018-08-02:91  "
## [60] "2018-07-05:91  "
## [61] "1st Qu.:0.0000  "
## [62] "0b274782-8e52-4f6a-bb17-36daa821f929: 1  "
## [63] "0ae1c621-387e-42a9-bcf3-7ad1c9b97ab4: 1  "
## [64] "0a6cae78-ea42-4e68-98c6-9d929068a38a: 1  "
## [65] "07780a1e-8af9-4b8a-bb9b-be8add15a1e0: 1  "
## [66] "06789d7b-b742-41d9-8556-79d23c193dc0: 1  "
## [67] "028eea3d-5c20-4afc-bb7e-a05bab305152: 1  "
## [68] "(Other) :81  "
## [69] "(Other)      :81  "
## [70] "(Other)      :33  "
## [71] "(Other)              :81  "
## [72] "(Other)              :129  "
## [73] "(Other)              :176  "
## [74] "(Other)              :182  "
```

```
# Create a summary of the "Effect" column
effect_summary <- summary(Neonics$Effect)
effect_summary <- summary(Litter$Effect)
sort(summary(Neonics$Effect), decreasing = TRUE)
```

```
##      Population      Mortality      Behavior Feeding behavior
##      1803          1493          360          255
##      Reproduction      Development      Avoidance      Genetics
##      197              136              102              82
##      Enzyme(s)          Growth          Morphology      Immunological
##      62                38                22              16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##      12                12                11              9
##      Physiology        Histology          Hormone(s)
##      7                  5                  1
```

Answer: Based on the result, the most common effects that studied are population, mortality, feeding behavior, reproduction and development

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 6)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##      667          285          183
##      Carniolan Honey Bee      Bumble Bee      (Other)
##      152          140          3196
```

```
species_summary <- summary(Neonics$Species.Common.Name)
species_table <- tibble::enframe(species_summary, name = "Species", value = "Count")

species_table <- species_table %>%
  arrange(desc(Count))

print(species_table)
```

```
## # A tibble: 100 x 2
##   Species      Count
##   <chr>      <int>
## 1 (Other)      670
## 2 Honey Bee    667
## 3 Parasitic Wasp 285
## 4 Buff Tailed Bumblebee 183
## 5 Carniolan Honey Bee 152
## 6 Bumble Bee   140
## 7 Italian Honeybee 113
## 8 Japanese Beetle 94
## 9 Asian Lady Beetle 76
## 10 Euonymus Scale 75
## # i 90 more rows
```

Answer: The six most common studied species in the dataset are honey bee, parasitic wasp, buff tailed bumblebee, carniolan honey bee, bumble bee, Italian honey bee. Most of this top 6 species are important pollinators.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

```
head(Neonics$Conc.1..Author.)
```

```
## [1] 27.2 19.7 47 25 13 268
## 1006 Levels: <0.0004 <0.025 <0.088 <0.5 <1.5 <10/ <2.5/ <4.00 <5.00 ... NR/
```

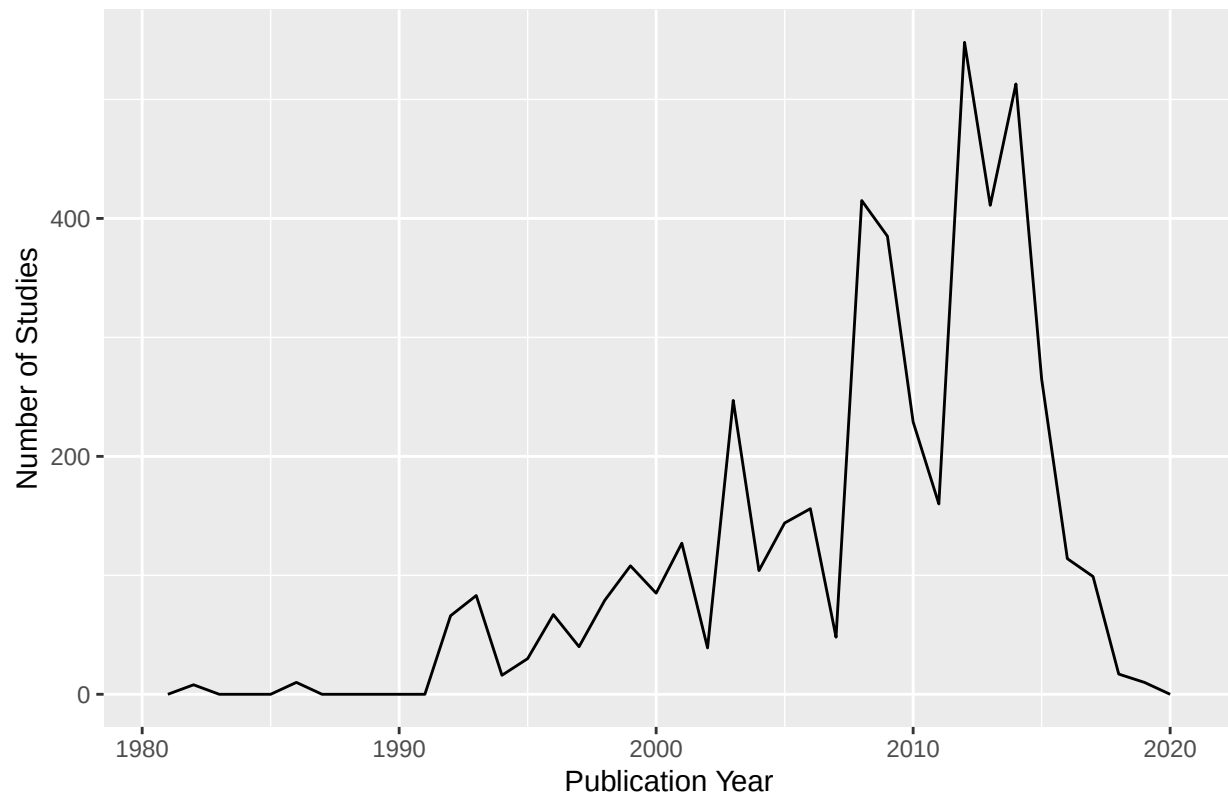
Answer: The class is Factor, and its because the columns contains non-numeric characters and texts.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

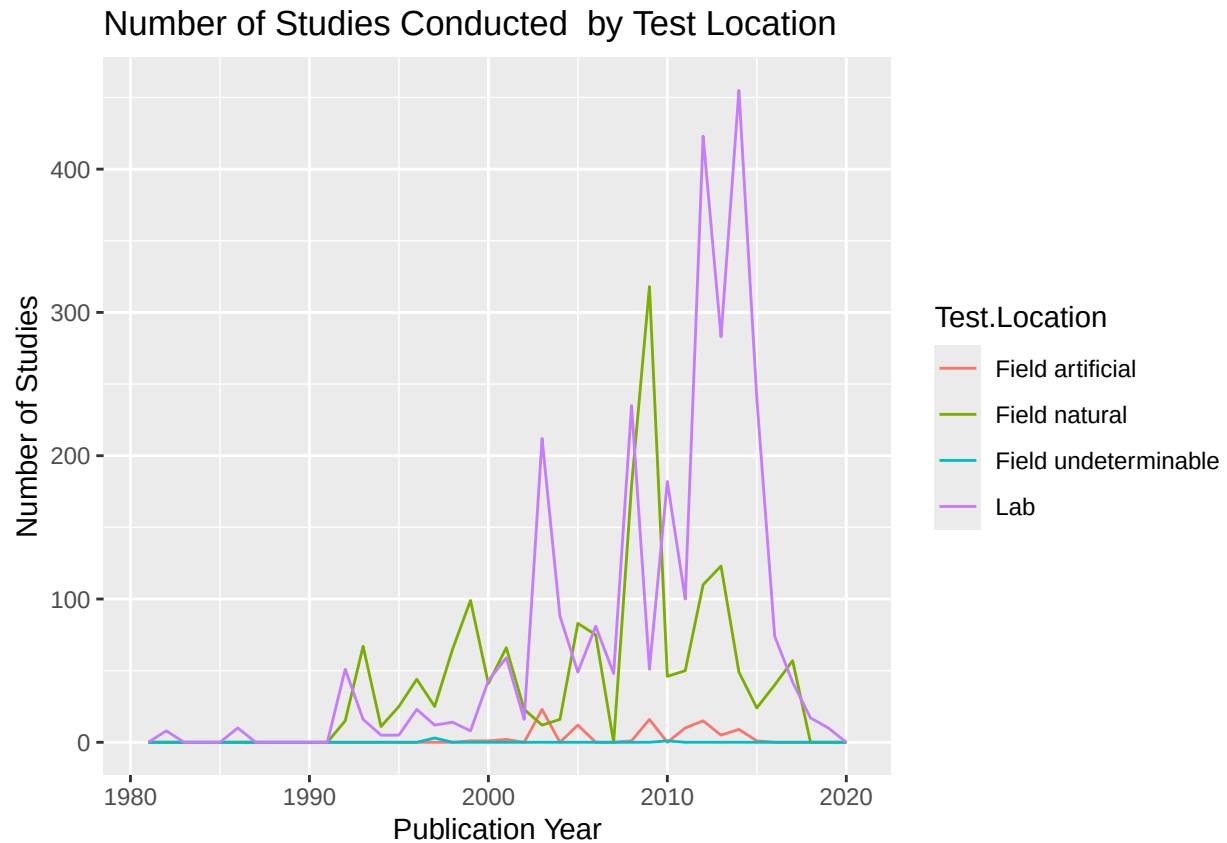
```
ggplot(data = Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1) +
  labs(title = "Number of Studies Conducted by Publication Year",
       x = "Publication Year", y = "Number of Studies")
```

Number of Studies Conducted by Publication Year



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(data = Neonics, aes(x = Publication.Year, color = Test.Location)) +  
  geom_freqpoly(binwidth = 1) +  
  labs(title = "Number of Studies Conducted by Test Location",  
        x = "Publication Year", y = "Number of Studies")
```



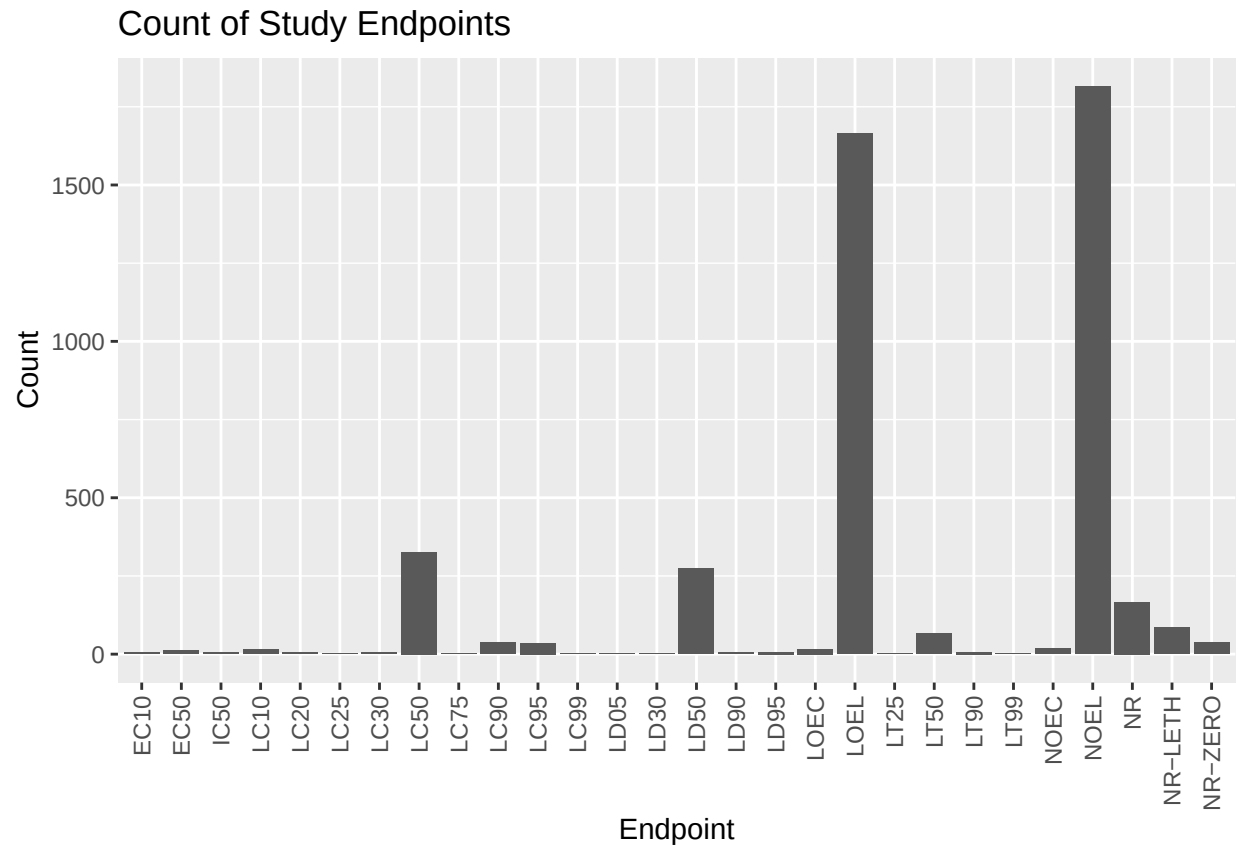
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are in Lab, followed by field natural. Lab location is increasing in trends by decades, and the field slowly increasing over the years with a big jump in the middle 2000s but then dropping down in the middle of 2010 and 2020. I assume mostly due to funding or COVID-19 influence.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(data = Neonics, aes(x = Endpoint)) +
  geom_bar() +
  labs(title = "Count of Study Endpoints", x = "Endpoint", y = "Count") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two common end points are LOEL and NOEL. Unless I misunderstood the question but the graph from my plot shows those two are with the highest. However, LC50 and LD50 are both also high end points.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- ymd(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# Find the unique dates
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

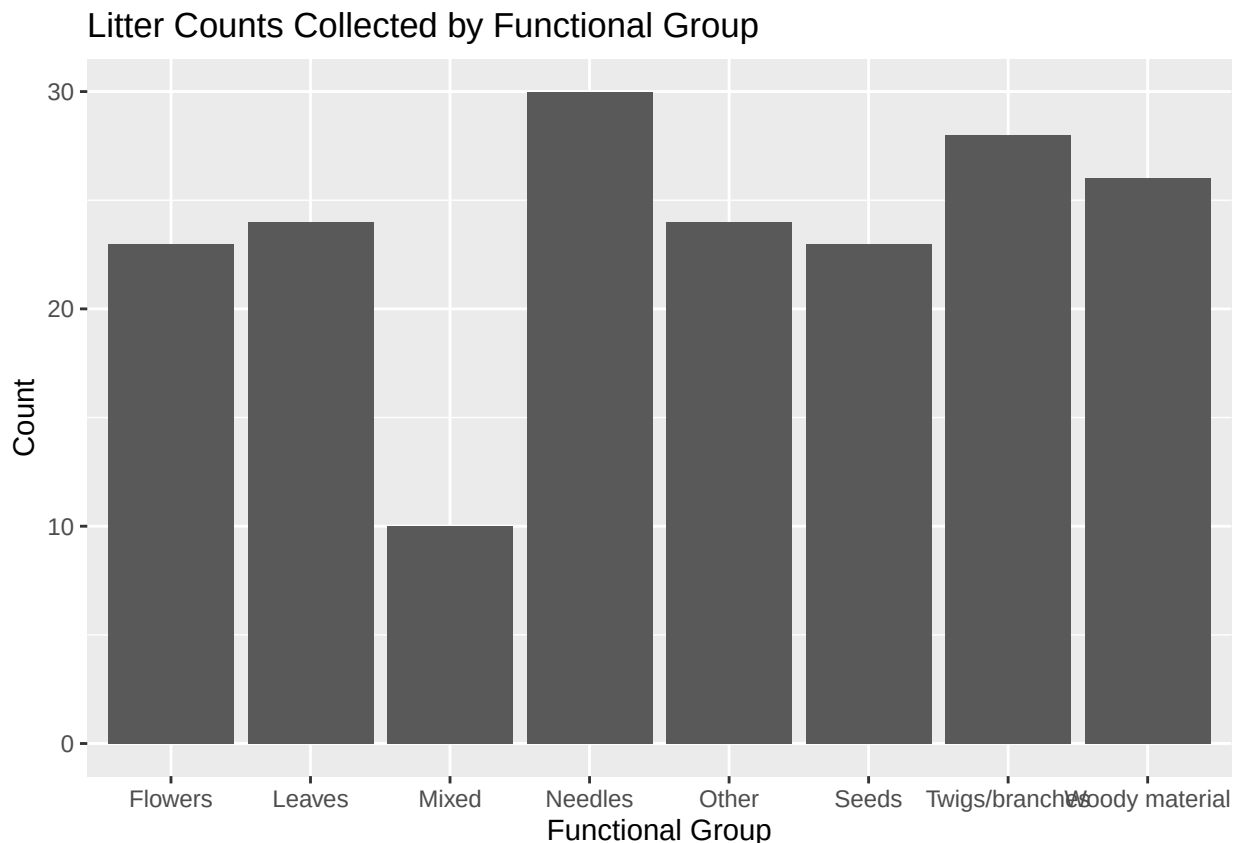
```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: The `summary` function returns frequency in a table showing the most common plot IDs, while the `unique` function returns vector listings plot IDs that shows in the data.

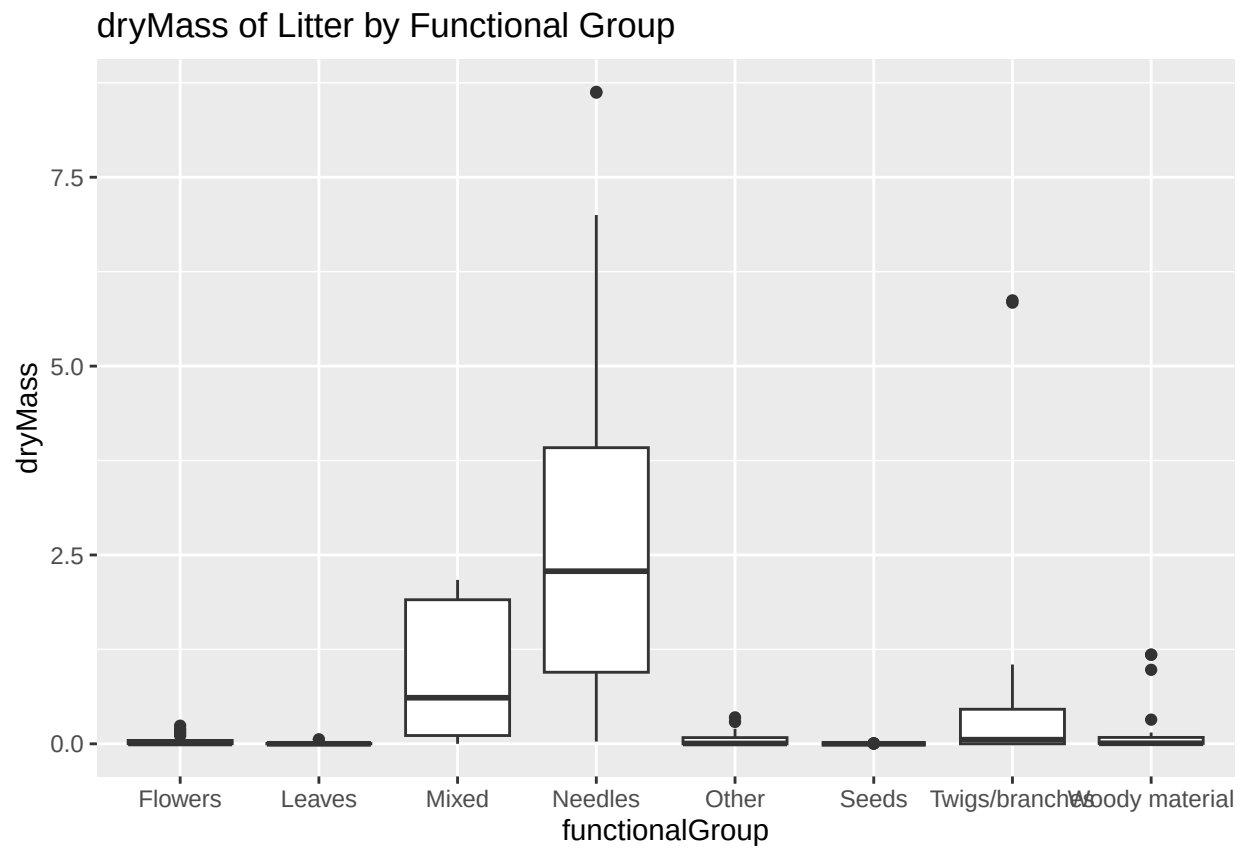
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(data = Litter, aes(x = functionalGroup)) +  
  geom_bar() +  
  labs(title = "Litter Counts Collected by Functional Group",  
        x = "Functional Group", y = "Count")
```

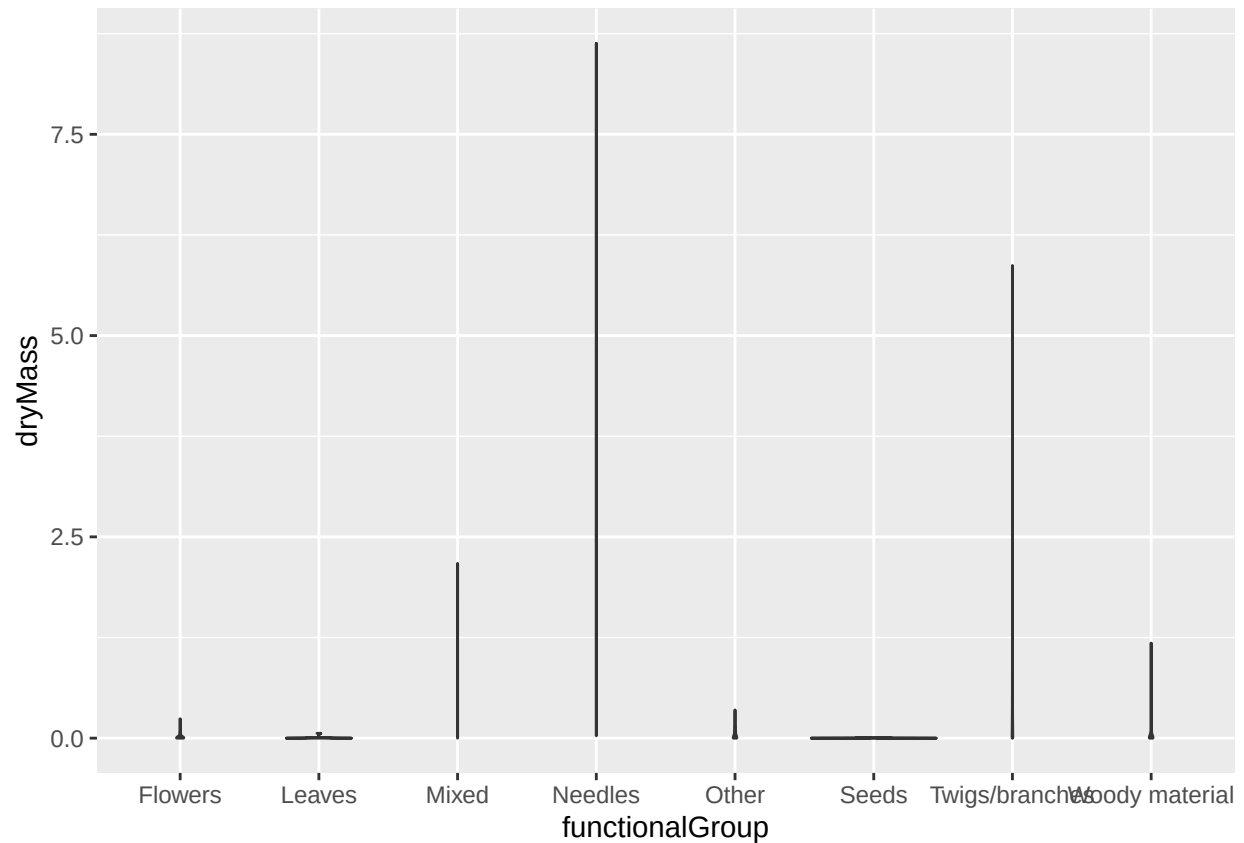


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
ggplot(data = Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_boxplot() +  
  labs(title = "dryMass of Litter by Functional Group")
```



```
# Violin plot  
ggplot(data = Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: Boxplot is easy to read and it shows the distribution clearly by showing median. Violin plot is visually not effective and might be hard to interpret by general audience and is less informative.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles have the highest biomass and follow by twigs/branch.